

DevOps Lab Program- 7

Name: Aditi Narasimhan

USN: 1RV24MC005

Project Structure:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ tree
.
└─ nginx-deployment.yaml

1 directory, 1 file
```

Steps for execution:

Step 1: Run *minikube start* to start minikube

```
1rv24mc005_aditi@aditi-Inspiron-3584:~$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...

🚨 Docker is nearly out of disk space, which may cause deployments to fail! (87% of capacity). You can pass '--force' to skip this check.
💡 Suggestion:

    Try one or more of the following to free up space on the device:

    1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
    2. Increase the storage allocated to Docker for Desktop by clicking on:
       Docker icon > Preferences > Resources > Disk Image Size
    3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
    📖 Related issue: https://github.com/kubernetes/minikube/issues/9024

🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: storage-provisioner, default-storageclass

❗ /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
   ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Check the status of minikube using *minikube status*

```
1rv24mc005_aditi@aditi-Inspiron-3584:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3:

Create a YAML file called nginx-deployment.yaml:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 4:

Apply the deployment file and verify whether it has been applied:

kubectl apply -f nginx-deployment.yaml

kubectl get deployments

kubectl get pods

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 0/2     2            0           18s
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 0/2     2            0           28s
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 0/2     2            0           49s
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment 2/2     2            2           2m18s
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5gh28 1/1     Running   0           5m22s
nginx-deployment-6f9664446b-fcph9 1/1     Running   0           5m22s
```

Step 5:

Expose the deployment as a service and verify:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl get svc
NAME          TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes    ClusterIP     10.96.0.1    <none>        443/TCP          40m
nginx-deployment NodePort      10.97.218.45 <none>        80:31689/TCP     9s
```

Step 6:

Run the service using minikube (access the web app):

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31689

```
Opening service default/nginx-deployment in default browser...
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ Gtk-Message: 00:21:05.366: Not loading module "atk-b
ridge": The functionality is provided by GTK natively. Please try to not load it.
```

Output:



Not Secure http://192.168.49.2:31689

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Step 7:

Delete all pods, services and deployments:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-5gh28" deleted
pod "nginx-deployment-6f9664446b-fcph9" deleted
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl delete services --all
service "kubernetes" deleted
service "nginx-deployment" deleted
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-7$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted
```